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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The notation of the blowup and the distinguished triangle attached to it. More on Morphisms, label more-morphisms-lemma-blowup (cited at statement line 3189) and Same chapter derham.tex, label lemma-distinguished-triangle-blowup (invoked at proof line 3196). The first fixes the blowup diagram $b : X' \to X$, $i : Z \to X$, $p : E \to Z$; the second supplies the triangle in the derived category of $(X \to S)^{-1}\mathcal{O}_S$-modules from which the two summands below arise.
- Splitting criterion in a triangulated category. Derived Categories, label derived-lemma-split (invoked at proof line 3203). It shows that a splitting of the map on the two summands whose image contains the summand $i_*\Omega^\bullet_{Z/S}$ yields a splitting of the first arrow of the triangle which vanishes on that summand.
- Decomposition into components of fixed relative dimension, and regularity of the closed immersion. Morphisms, label morphisms-lemma-smooth-omega-finite-locally-free (invoked at proof line 3214) and Divisors, label divisors-lemma-immersion-smooth-into-smooth-regular-immersion (invoked at proof line 3221). They reduce to X smooth of a fixed relative dimension N and show that each $i_m : Z_m \to X$ is a regular immersion of codimension $N - m$, so that the projective bundle formula applies to $E_m \to Z_m$.
- The projective space bundle formula and the description of its summands. Same chapter derham.tex, remark remark-projective-space-bundle-formula (invoked at proof lines 3235, 3251 and 3282). It supplies the isomorphism $\tilde\xi$ identifying $Rp_*\Omega^\bullet_{E/S}$ with a direct

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sum of shifts of $\Omega^\bullet_{Z/S}$, together with the subcomplexes M_m and K_m satisfying $M = K[-2]$ and the commutativity of the diagram involving θ'.
- The Gysin-type map of the blowup and its relation to the first Chern class of the exceptional line bundle. Same chapter derham.tex, label lemma-gysin-via-log-complex (invoked at proof line 3273) and remark remark-cup-product-as-a-map (invoked at proof line 3280). They provide the map $\delta : \Omega^\bullet_{E/S}[-2] \to \Omega^\bullet_{X'/S}$ whose composite with the restriction back to E is θ' for $c_1^{dR}(\mathcal{O}_E(1))$, which is what makes the displayed chain a section.

Source reference. Original lemma 05QT: sources/stacks-chow/tags.txt, source lines 7387–7387. Navigation pointer for \ref{derived-lemma-split} used inside the counted span (line 3203); the argument that consumes it is retained in the span.

Source reference. Original lemma 067U: sources/stacks-chow/tags.txt, source lines 7983–7983. Navigation pointer for \ref{divisors-lemma-immersion-smooth-into-smooth-regular-immersion} used inside the counted span (line 3221); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUV: sources/stacks-derham/pinned-original-source.tex, source lines 3132–3132. Navigation pointer for \ref{lemma-distinguished-triangle-blowup} used inside the counted span (line 3196); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUQ: sources/stacks-derham/pinned-original-source.tex, source lines 2560–2560. Navigation pointer for \ref{lemma-gysin-via-log-complex} used inside the counted span (line 3273); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FUT: sources/stacks-chow/tags.txt, source lines 19788–19788. Navigation pointer for \ref{more-morphisms-lemma-blowup} used inside the counted span (line 3189); the argument that consumes it is retained in the span.

Source reference. Original lemma 02G1: sources/stacks-chow/tags.txt, source lines 3212–3212. Navigation pointer for \ref{morphisms-lemma-smooth-omega-finite-locally-free} used inside the counted span (line 3214); the argument that consumes it is retained in the span.

Source reference. Original remark 0FU7: sources/stacks-derham/pinned-original-source.tex, source lines 575–575. Navigation pointer for \ref{remark-cup-product-as-a-map} used inside the counted span (line 3280); the argument that consumes it is retained in the span.

Source reference. Original remark 0FUN: sources/stacks-derham/pinned-original-source.tex, source lines 2278–2278. Navigation pointer for \ref{remark-projective-space-bundle-formula} used inside the counted span (line 3235); the argument that consumes it is retained in the span.

Proposition 0.1. *With notation as in More on Morphisms, Lemma 0FUT the map $\Omega^\bullet_{X/S} \to Rb_*\Omega^\bullet_{X'/S}$ has a splitting in $D(X, (X \to S)^{-1}\mathcal{O}_S)$.*

Proof. Consider the triangle constructed in Lemma 0FUV. We claim that the map

$$Rb_*(\Omega^\bullet_{X'/S}) \oplus i_*\Omega^\bullet_{Z/S} \to i_*Rp_*(\Omega^\bullet_{E/S})$$

has a splitting whose image contains the summand $i_*\Omega^\bullet_{Z/S}$. By Derived Categories, Lemma 05QT this will show that the first arrow of the triangle has a splitting which

vanishes on the summand $i_*\Omega^\bullet_{Z/S}$ which proves the lemma. We will prove the claim by decomposing $Rp_*\Omega^\bullet_{E/S}$ into a direct sum where the first piece corresponds to $\Omega^\bullet_{Z/S}$ and the second piece can be lifted through $Rb_*\Omega^\bullet_{X'/S}$.

Proof of the claim. We may decompose X into open and closed subschemes having fixed relative dimension to S, see Morphisms, Lemma 02G1. Since the derived category $D(X, f^{-1}\mathcal{O})_S)$ correspondingly decomposes as a product of categories, we may assume X has fixed relative dimension N over S. We may decompose $Z = \coprod Z_m$ into open and closed subschemes of relative dimension $m \geq 0$ over S. The restriction $i_m : Z_m \to X$ of i to Z_m is a regular immersion of codimension $N - m$, see Divisors, Lemma 067U. Let $E = \coprod E_m$ be the corresponding decomposition, i.e., we set $E_m = p^{-1}(Z_m)$. If $p_m : E_m \to Z_m$ denotes the restriction of p to E_m, then we have a canonical isomorphism

$$\tilde{\xi}_m : \bigoplus\nolimits_{t=0,\ldots,N-m-1} \Omega^\bullet_{Z_m/S}[-2t] \longrightarrow Rp_{m,*}\Omega^\bullet_{E_m/S}$$

in $D(Z_m, (Z_m \to S)^{-1}\mathcal{O}_S)$ where in degree 0 we have the canonical map $\Omega^\bullet_{Z_m/S} \to Rp_{m,*}\Omega^\bullet_{E_m/S}$. See Remark 0FUN. Thus we have an isomorphism

$$\tilde{\xi} : \bigoplus\nolimits_m \bigoplus\nolimits_{t=0,\ldots,N-m-1} \Omega^\bullet_{Z_m/S}[-2t] \longrightarrow Rp_*(\Omega^\bullet_{E/S})$$

in $D(Z, (Z \to S)^{-1}\mathcal{O}_S)$ whose restriction to the summand $\Omega^\bullet_{Z/S} = \bigoplus \Omega^\bullet_{Z_m/S}$ of the source is the canonical map $\Omega^\bullet_{Z/S} \to Rp_*(\Omega^\bullet_{E/S})$. Consider the subcomplexes M_m and K_m of the complex $\bigoplus_{t=0,\ldots,N-m-1} \Omega^\bullet_{Z_m/S}[-2t]$ introduced in Remark 0FUN. We set

$$M = \bigoplus M_m \quad \text{and} \quad K = \bigoplus K_m$$

We have $M = K[-2]$ and by construction the map

$$c_{E/Z} \oplus \tilde{\xi}|_M : \Omega^\bullet_{Z/S} \oplus M \longrightarrow Rp_*(\Omega^\bullet_{E/S})$$

is an isomorphism (see remark referenced above).

Consider the map

$$\delta : \Omega^\bullet_{E/S}[-2] \longrightarrow \Omega^\bullet_{X'/S}$$

in $D(X', (X' \to S)^{-1}\mathcal{O}_S)$ of Lemma 0FUQ with the property that the composition

$$\Omega^\bullet_{E/S}[-2] \longrightarrow \Omega^\bullet_{X'/S} \longrightarrow \Omega^\bullet_{E/S}$$

is the map θ' of Remark 0FU7 for $c_1^{dR}(\mathcal{O}_{X'}(-E))|_E) = c_1^{dR}(\mathcal{O}_E(1))$. The final assertion of Remark 0FUN tells us that the diagram

$$\begin{array}{ccc} K[-2] & \xrightarrow{\quad\quad} & M \\ {\scriptstyle (\tilde{\xi}|_K)[-2]}\downarrow & & \downarrow{\scriptstyle \tilde{x}|_M} \\ Rp_*\Omega^\bullet_{E/S}[-2] & \xrightarrow{Rp_*\theta'} & Rp_*\Omega^\bullet_{E/S} \end{array}$$

commutes. Thus we see that we can obtain the desired splitting of the claim as the map

$$\begin{aligned} Rp_*(\Omega^\bullet_{E/S}) &\xrightarrow{(c_{E/Z} \oplus \tilde{\xi}|_M)^{-1}} \Omega^\bullet_{Z/S} \oplus M \\ &\xrightarrow{\mathrm{id} \oplus \mathrm{id}^{-1}} \Omega^\bullet_{Z/S} \oplus K[-2] \\ &\xrightarrow{\mathrm{id} \oplus (\tilde{\xi}|_K)[-2]} \Omega^\bullet_{Z/S} \oplus Rp_*\Omega^\bullet_{E/S}[-2] \\ &\xrightarrow{\mathrm{id} \oplus Rb_*\delta} \Omega^\bullet_{Z/S} \oplus Rb_*\Omega^\bullet_{X'/S} \end{aligned}$$

The relationship between θ' and δ stated above together with the commutative diagram involving θ', $\tilde{\xi}|_K$, and $\tilde{\xi}|_M$ above are exactly what's needed to show that this is a section to the canonical map $\Omega^\bullet_{Z/S} \oplus Rb_*(\Omega^\bullet_{X'/S}) \to Rp_*(\Omega^\bullet_{E/S})$ and the proof of the claim is complete. $\square$